

Python Programming

Assignment 1

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1 Theory

1.1 Variables

- **What is it?** A variable is a name we give to a value so that we can use it later in the program. In Python we don't have to tell the type of variable before hand, Python decide it by itself when we give the value.
- **Why do we need it?** We need variable because we cannot keep writing same value again and again. If value change at one place we don't have to change at every place, we just change the variable value.
- **Where and How is it used?** It is used almost everywhere in program, like storing user input, storing result of calculation, counters in loop etc. Example: `age = 20`, here `age` is variable and 20 is value stored in it.

1.2 Basic Data Types

1.2.1 Integer

- **What is it?** Integer is whole number, without decimal point, like 5, 100, -3.
- **Why do we need it?** We need it for counting things, like number of students, marks etc.
- **Where and How is it used?** In Python we just write `x = 10` and it become integer automatically.

1.2.2 Float

- **What is it?** Float is number which have decimal point, like 3.14, 2.5.
- **Why do we need it?** Needed when we deal with measurement or values which are not whole number, like price, height, average marks.
- **Where and How is it used?** Written as `y = 3.14`.

1.2.3 Boolean

- **What is it?** Boolean have only two value, `True` or `False`.
- **Why do we need it?** It is used to check condition, like is something correct or not.
- **Where and How is it used?** Example `is_pass = True`. Mostly used in if condition and loops.

1.2.4 String

- **What is it?** String is sequence of character, used to store text.
- **Why do we need it?** Needed to store name, address, message, basically any text data.

- **Where and How is it used?** Written inside single or double quote, like `name = "Kunal"`.

1.3 Arithmetic Operators

- **What is it?** These are operator used to do maths calculation like addition, subtraction, multiplication, division. Example: `+`, `-`, `*`, `/`, `%`, `**`, `//`.
- **Why do we need it?** Every program need some calculation at some point, so this operator help to do that calculation directly in code.
- **Where and How is it used?** Used in billing system, calculator app, average calculation etc. Example: `total = price * quantity`.

1.4 Comparison Operators

- **What is it?** Comparison operator is used to compare two value, result is always boolean (True/False). Example: `==`, `!=`, `>`, `<`, `>=`, `<=`.
- **Why do we need it?** We need it to make decision in program, like checking if one number is bigger then other, or checking password is correct or not.
- **Where and How is it used?** Mostly used inside `if` statement. Example: `if age >= 18:`.

1.5 Assignment Operators

- **What is it?** Assignment operator is used to give value to variable. Simple one is `=`, but there are more like `+=`, `-=`, `*=`, `/=` which update the value directly.
- **Why do we need it?** It make our code short. Instead of writing `x = x + 1` we can just write `x += 1`.
- **Where and How is it used?** Used when we need to update value of variable again and again, like counter in loop. Example: `count += 1`.

1.6 Type Conversion

- **What is it?** Type conversion mean changing one data type into another, like converting string into integer or integer into float.
- **Why do we need it?** Because sometime the data we get is in wrong type for our need, for example `input()` always give string, but if we want to do maths on it we have to convert it to integer first.
- **Where and How is it used?** Used with function like `int()`, `float()`, `str()`. Example: `age = int(input("Enter age: "))`.

1.7 Input and Output

1.7.1 `input()`

- **What is it?** This function is used to take input from user while program is running.
- **Why do we need it?** Program become interactive, user can give their own data instead of fix value written in code.
- **Where and How is it used?** `name = input("Enter your name: ")`. Note that whatever user type, it is always stored as string.

1.7.2 `print()`

- **What is it?** This function is used to show output on screen.
- **Why do we need it?** Without this we cannot see result of our program.
- **Where and How is it used?** `print("Hello World")` or `print(variable_name)`.

1.8 Strings

1.8.1 String Creation

- **What is it?** String created by putting text inside single quote or double quote. Example: `s = "Python"`.

1.8.2 Indexing

- **What is it?** Every character in string have a position number called index, starting from 0. We can access single character using this index, like `s[0]` give first character. Negative index like `s[-1]` give last character.

1.8.3 Slicing

- **What is it?** Slicing is used to take a part of string, not just single character. Syntax is `s[start:end]`, it give character from start index to end-1 index.

1.8.4 String Operations

- **What is it?** Some common operation are concatenation (joining two string using `+`), repetition (`*`), and checking length using `len()`. String are used everywhere, like storing name, printing message, processing text file etc.

1.9 Lists

1.9.1 Creating Lists

- **What is it?** List is collection of item, written inside square bracket, separated by comma. Example: `fruits = ["Apple", "Banana", "Mango"]`. List can store different type of data together also.

1.9.2 Indexing

- **What is it?** Same like string, each item in list has index starting from 0, we can access item using that index. Example `fruits[0]` give "Apple".

1.9.3 Slicing

- **What is it?** We can take part of list using slicing, same syntax as string, `fruits[1:3]`.

1.9.4 Updating Elements

- **What is it?** List is mutable, means we can change value of any element directly using index. Example: `fruits[0] = "Grapes"`.

1.9.5 Nested Lists

- **What is it?** A list can contain another list inside it, this is called nested list. Example: `a = [[1,2],[3,4]]`. This is useful to represent table like data, matrix etc.

1.9.6 Basic List Methods

- `append()` - add new element at end of list.
- `insert()` - add element at particular position.
- `remove()` - remove particular value from list.
- `pop()` - remove element from end (or given index) and also return it.
- `sort()` - arrange list in increasing order.
- `reverse()` - reverse the order of list.
- `len()` - give total number of element present in list.

We need list because in real life we always deal with collection of data, not single value, like list of students, list of marks etc. List give easy way to store and manage this kind of data together.

2 Assignment 2

Aim

Write code using given Datatypes: List, Tuple, Set, Dictionary and String, with one simple example each.

2.1 List

- **What is it?** List is a data structure that store many value together in order, written inside square bracket [].
- **Why do we need it?** It help us to keep related item together in one place and we can change them later whenever needed.
- **Where and How is it used?** Used for storing marks, names, shopping item list, or any group of data which keep changing.

Simple Code

```
1 marks = [80, 70, 95]
2 marks.append(60)
3 print(marks)
```

Output: [80, 70, 95, 60]

2.2 Tuple

- **What is it?** Tuple is almost same as list but it is written with round bracket (), and once it is made we cannot change value inside it.
- **Why do we need it?** It keep our data safe, so it don't get change by mistake in program.
- **Where and How is it used?** Used for fixed kind of data, like coordinate (x, y) or date (day, month, year).

Simple Code

```
1 point = (2, 3)
2 x, y = point
3 print(x, y)
```

Output: 2 3

2.3 Set

- **What is it?** Set is data structure which store value with no fix order and it don't allow repeat value, written with curly bracket { }.
- **Why do we need it?** It automatically remove duplicate value and also check if item is present very fast.

- **Where and How is it used?** Used to find unique item, like unique visitor on website or unique word in some text.

Simple Code

```
1 nums = [1, 2, 2, 3, 3, 3]
2 unique = set(nums)
3 print(unique)
```

Output: {1, 2, 3}

2.4 Dictionary

- **What is it?** Dictionary store data as key and value pair, written like {key: value}.
- **Why do we need it?** It let us find value fast by using its key, we don't have to search one by one like in list.
- **Where and How is it used?** Used for storing detail like student name and marks together, or setting of some app.

Simple Code

```
1 student = {"name": "Alice", "marks": 90}
2 print(student["name"])
```

Output: Alice

2.5 String

- **What is it?** String is sequence of character, written inside quote.
- **Why do we need it?** It let us store and work with text data, like name or sentence.
- **Where and How is it used?** Used for text related work, such as taking input, printing message, and processing word.

Simple Code

```
1 text = "hello"
2 print(text.upper())
```

Output: HELLO

2.6 Quick Comparison

Data Structure	Brackets	Ordered	Can Change	Allows Repeats
List	[]	Yes	Yes	Yes
Tuple	()	Yes	No	Yes
Set	{ }	No	Yes	No
Dictionary	{k:v}	Yes	Yes	Keys: No, Values: Yes
String	" "	Yes	No	Yes